Review for STAT301 Midterm

Midterm information:

- The STAT301 midterm exam will be on **Thursday, July 23, 2026, 3:45pm–4:45pm** (60 minutes) in lecture room **(ESB1012), in-person**.
- Hard copies of the exam will be distributed. You will have 60 minutes for writing your solutions, at most 15 minutes (4:45pm – 5:00pm) for uploading your solutions in a **single pdf file** to Canvas "**Assignment**" section (click "**Midterm Solutions upload here**" link).
- You can either write your solutions on the exam paper, scan it, then upload to Canvas, or write on your laptop and then upload to Canvas.

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- The midterm exam is **closed books/notes**. You may bring in **one letter-size sheets of notes (both sides, written or typed)**.
- You are allowed to access Canvas/Internet only at the end of the exam for uploading your solutions.
- There will be some simple calculations, so you need a **simple calculator**.
- Please make sure you **write clearly** (easy to read by graders), or you may lose points.
- Please see more details on Canvas.

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The midterm exam will focus on **understanding of the basics of linear regression models**, and **critical thinking**. Not tedious computations or mathematical arguments.

For example, how to interpret computer R outputs, what do different components of the model mean, and explain certain concepts in the given context (dataset).

R code will not be directly tested, but you need to understand the outputs of R code.

Since a simple linear regression model is a special case of multiple linear regression models when the number of predictors is one, in this review we will focus on multiple linear regression models.

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A multiple linear regression model can be written as

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + e,$$

where the β_j's are unknown parameters (to be estimated from data), e is random error. The above model is also called an **additive model**.

Given a dataset (a sample of size n), the above model can also be written as

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_p x_{pi} + e_i, \qquad i = 1, 2, \cdots, n.$$

In a linear regression model, the the response y must be a continuous variable, while the predictors X_j's can be either continuous or categorical.

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The **assumptions** for a linear regression model are

- The relationship between the response and the predictors is **linear** (in the parameters), or $E(e) = 0$.
- The response data $y_1, y_2, \cdots, y_n$ (or the errors $e_1, e_2, \cdots, e_n$) are **independent**.
- The variance of the y_i's (or the e_i's) are **constant**.
- The responses $y_1, y_2, \cdots, y_n$ (or the errors $e_1, e_2, \cdots, e_n$) follow a **normal distribution**, i.e., $e_i \sim N(0, \sigma^2)$.

Violation of any of these assumptions in data analysis may lead to misleading results/conclusions.

Note that, in a linear regression model, the predictors x_j's and the parameters β_j's are assumed **fixed** (i.e., not random variables), while the response y and the error e are assumed to be random variables.

This is not the case in a Bayesian framework.

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The unknown parameters β_j's are estimated based on the **least square method**, which finds values of the β_j's to minimize the following sum of squares

$$\sum_{i=1}^{n}(y_i - (\beta_0 + \beta_1 x_i + \cdots + \beta_p x_p))^2.$$

The resulting estimates are denoted by $\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_p$ (called least square estimates of the parameters).

The **fitted values** of the response data are

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i + \cdots + \hat{\beta}_p x_p, \qquad i = 1, 2, \cdots, n.$$

The **residuals** are

$$r_i = y_i - \hat{y}_i, \qquad i = 1, 2, \cdots, n.$$

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For a multiple linear regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p + e,$$

the interpretation of the parameters are as follows.

If (say) x_1 is a **continuous variable**, one unit change in x_1 is associated with β_1 units change in y, holding other predictors fixed (or adjusted for other predictors, or incorporating other predictors).

In a simple linear regression case, β_1 represents the slope (or rate of change) of the fitted straight line, and β_0 represent the intercept of the line.

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If (say) x_2 is a **categorical variable** with q categories, computer will produce $q-1$ estimates, with the reference category (default is the first category) hidden, and the (say) j-th estimate is the change in y for category $j-1$ compared with the reference category ($j = 2, 3, \cdots, q$), holding other predictors fixed (or adjusted for other predictors, or incorporating other predictors).

In R, categorical variables in a linear regression model are coded based on **dummy variables.** So, x_2 is equivalent to $q-1$ (binary) dummy variables.

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A significant **interaction** term in the model, say $x_1 x_2$, suggests that the association between y and x_1 may depend on the value of x_2. In other words, without knowing the value of x_2, we don't know how change in x_1 may affect the value of y.

An interaction term makes the interpretation more complicated. Thus, if the interaction term is not statistically significant, we should remove it from the model.

An interaction may happen between any two predictors in the model.

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An interaction can be detected by testing a hypotheses on the parameter associated with the interaction term (e.g., a model may contain the term $\beta_3 x_1 x_2$).

To test the significance of an interaction term, we may perform a test for hypotheses

$$H_0 : \beta_3 = 0 \qquad \text{versus} \qquad \beta_3 \neq 0.$$

The test statistic is

$$z = \frac{\hat{\beta}_3}{SE(\hat{\beta}_3)}.$$

We then compute a p-value or decide reject/accept H_0 based on a standard normal distribution.

Note that the true value of β_3 is unknown (maybe never known), we estimate it based on the data. The estimate $\hat{\beta}_3$ may or may not be close to the true value. **Statistical inference for β_3 (or any other parameter) is to learn something about the true value.**

There are two ways to do statistical inference: construct a **confidence interval for β_3** or perform a **hypothesis testing for β_3**. These two methods are equivalent.

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"A 95% confidence interval for β_3 contains 0" is equivalent to "fail to reject $H_0 : \beta_3 = 0$ at 5% level". In this case, the true value of β_3 may be 0, so there may be no interaction.

Note that a t-distribution may be approximated by a normal distribution. So, for simplicity, we can just use normal distribution in statistical inference for linear regression models.

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Given a dataset, after we fit a linear regression model, we must check if the model assumptions hold or not (i.e., **model diagnostics**). Otherwise, the results may be misleading or biased.

Two common tools for model diagnostics:

- **residual plots:** plot residuals against fitted values. They can be used to check linearity and constant variance assumptions.
- **QQ plot**: plot residual quantiles against normal quantiles. It can be used to check normality assumption.

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If model diagnostics suggest that some assumptions may be violated, we can do the following to possibly fix the problems:

- **Variable transformations.** They can be used to possibly fix non-linearity, non-normality, and non-constant variance problems.
- **Adding additional terms in the model, such as interaction terms or quadratic terms**.
- **Bootstrap.** Useful for non-normal data or small sample sizes. It is often used to calculate more reliable standard errors.

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If the response data are **not independent** (e.g., repeated measurements, longitudinal data), we should not use the usual linear regression models. There are many different models/methods available for longitudinal data analysis.

Independence can be informally judged by (i) how the data are collected; and (ii) common sense.

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Another issue with a linear regression model is **multi-collinearity**. If the problem exists, the standard errors and p-values with the predictors may be inflated (i.e., larger than necessary).

Common tools to fix multi-collinearity problem

- **check pairwise correlations for continuous predictors.** If two predictors are highly correlated, maybe we can just use one of them or combine them.
- **variance inflation factor (VIF).** Check the values of VIF (or generalized VIF) to see if they are too large (compared with 5 or $\sqrt{5}$).

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When interpreting data analysis results, we should be careful about the difference between **association** and **causation**. They are often based on how the data are collected: observational study or designed experiments.

If the data are collected from an **observational study**, any significant predictors only suggest **associations** with the response, not causation.

If the data are collected from a **designed experiment involving randomization**, a significant predictor (e.g., treatment) may suggest a **causation** with the response.